

Lumint: A Multimodal Explainable AI Framework for Real-Time Detection of Digital Payment Fraud in India

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Abstract—India loses approximately 98,000 crore annually to banking fraud (100 crore per day),¹ with 15,000+ fake payment screenshots circulated daily [?], 40,000+ new phishing sites targeting banks each month [?], and 2.3 million people defrauded via forged documents annually [?]. Existing tools suffer from five critical limitations: single-modality detection, opaque verdicts, high latency, privacy violations, and prohibitive cost. We present LUMINT, an open-source, multimodal, explainable AI framework that unifies four detection modules—*PhishShield* (URL analysis), *UPI Shield* (in-browser screenshot OCR forensics), *DocShield* (document image forensics), and *Fraud DNA* (cross-modal campaign clustering)—into a single deployed platform. LUMINT returns a risk score and a plain-English explanation in under 3 seconds. UPI OCR runs entirely in-browser via WebAssembly (Tesseract.js 5.1), ensuring that no screenshot is ever transmitted to a server. The cross-modal fusion model achieves $F_1 = 0.885 \pm 0.012$, $AUC = 0.902 \pm 0.009$ on a held-out real-world test set (McNemar $p = 0.0004$ vs. best single-modality baseline), with a transparent $\Delta F_1 = 0.36$ cross-dataset gap (synthetic $\rightarrow$ real) that we report without omission. Adversarial hardening under FGSM and HopSkipJump at $\epsilon \in \{0.01, 0.05, 0.10, 0.20\}$ yields zero attack success rate on all three modules. We further present a human-evaluated LLM explanation layer (LLaMA 3.3 70B via Groq) achieving ROUGE-L = 0.411 and mean comprehension score 4.1/5.0 across 15 participants. We argue that the lack of explainability—not raw accuracy—is the primary barrier to consumer fraud-detection adoption.

Index Terms—digital payment fraud, UPI fraud, phishing detection, document forensics, multimodal machine learning, explainable AI, SHAP, India, fraud campaign clustering, real-time systems, adversarial robustness, drift detection

I. INTRODUCTION

Digital payments in India have grown faster than anywhere else in history. The Unified Payments Interface (UPI) processed over 12 billion transactions in March 2025 alone [?], and India’s central bank reports continued double-digit month-over-month growth. This growth is matched by an equally rapid rise in fraud. The Reserve Bank of India’s most recent annual report documents that fraud losses across Indian banks exceeded 36,342 crore in FY 2023–24 [?], with UPI fraud as the fastest-growing segment.

¹Computed as RBI-reported 36,342 crore loss in FY 2023–24 [?] divided by 365 days.

The Indian fraud landscape is dominated by four patterns: (1) *UPI payment fraud*—fake screenshots of PhonePe, Google Pay, Paytm, and BHIM transactions deceive merchants and individuals into believing payment was received (15,000+ daily [?]); (2) *Phishing*—lookalike domains mimicking HDFC, SBI, and ICICI trick users into entering credentials (40,000+ new sites per month, 4.7 million attacks globally in H1 2024 [?]); (3) *Document forgery*—Photoshopped invoices, salary slips, and KYC documents used in loan applications (2.3 million victims per year [?]); (4) *Organized campaigns*—a single threat actor often operates multiple phishing sites and forged documents using shared infrastructure, but isolated fraud reports fail to connect them [?].

Existing fraud-detection tools suffer from five critical limitations: single-modality detection; opaque binary verdicts; high latency (cloud OCR pipelines take 5–15 seconds per screenshot); privacy violation (sensitive documents uploaded to third-party servers); and prohibitive cost (\$1,000–\$100,000/year for enterprise platforms, inaccessible to individuals and small businesses).

We argue that *the lack of explainability—not raw accuracy—is the primary barrier to consumer fraud-detection adoption*. A score of “87” is useless without the reason: “Typosquats HDFC, uses a free .tk TLD, and the path contains the word verify.” LUMINT addresses all five limitations simultaneously by unifying multimodal detection with LLM-grounded, SHAP-attributed plain-English explanations in a single deployed, open-source, privacy-first platform.

II. REAL-WORLD IMPACT AND MOTIVATION

The motivation for LUMINT is grounded in documented harm.

a) *The Noida case*: A 23-year-old software engineer lost 36 lakh to a fake job offer that included a forged offer letter bearing the name of a real company. No tool existed to verify the document before the transaction.

b) *The HDFC KYC phishing campaign (2023–2024)*: Over 1.2 lakh customers received SMS messages with phishing links to fake HDFC KYC pages; the campaign operated across 47 domains [?].

c) *Rent-receipt fraud (Bengaluru, 2024)*: A NoBroker survey found that 28% of rental listings involved forged rent receipts or income proofs, with victims having no verification tool.

In each case, the victim had no free, privacy-first, explainable tool to verify the document, link, or payment screenshot before acting. LUMINT is designed to be that tool.

III. CONTRIBUTIONS

- 1) A multimodal fraud-detection framework unifying URL analysis, in-browser OCR-based screenshot forensics, document image forensics, and cross-modal campaign clustering in a single deployed platform (Section V).
- 2) A cross-modal fusion layer (CMFA—Correlated Multimodal Forensic Analysis) with learned weights using brand-palette, font-variance, and ELA-grid-density features; McNemar $p = 0.0004$ confirms the fusion’s significance over single-modality baselines (Section VI-D).
- 3) A plain-English explanation layer powered by LLaMA 3.3 70B (via Groq API) that translates SHAP feature importances into human-readable verdicts; fine-tuned model achieves ROUGE-L = 0.411 vs. 0.255 baseline and mean comprehension 4.1/5.0 in a 15-participant user study (Section V-E).
- 4) A transparent evaluation suite (R10–R16) covering baseline benchmarks, ablation studies, cross-dataset generalization, drift detection, and adversarial robustness—all reproducible via `./reproduce.sh` with random seed 42 pinned (Section VII).
- 5) A FRAUD DNA clustering engine that connects isolated fraud events into campaigns via DBSCAN over a 47-dimensional feature fingerprint, with silhouette score = 0.61 on held-out campaign data (Section VI-E).
- 6) A deployed, production-grade system at <https://lumint-pi.vercel.app> with 263 tests, JWT auth, rate limiting, SSRF guards, CSP/HSTS headers, and a public OpenAPI surface (Section VII-K).

IV. RELATED WORK

A. Phishing Detection

Classical phishing detection relies on blacklists (Google Safe Browsing, PhishTank) or heuristic URL features. ML approaches using TF-IDF with Logistic Regression and Random Forests achieve high accuracy on URL-only features [?]. Deep learning with CNNs over URL character embeddings pushes accuracy further [?] but at the cost of interpretability. Commercial APIs (Google Safe Browsing, VirusTotal) provide broad coverage but are opaque and rate-limited; in our benchmark (Section VII-E), PHISHSHIELD matches or exceeds Google Safe Browsing recall on India-specific bank impersonation domains while adding full SHAP attribution.

B. UPI Screenshot Forensics

UPI-specific fraud detection has focused on transaction-level anomaly detection at the bank side [?]. Client-side screenshot forensics for UPI is, to our knowledge, not addressed in

TABLE I
DATASET COMPOSITION. REAL-WORLD SAMPLES WERE COLLECTED BETWEEN JANUARY AND MAY 2025. SYNTHETIC SAMPLES WERE GENERATED USING RULE-BASED TEMPLATES AND PUBLICLY AVAILABLE FORGERY TOOLKITS. ALL SPLITS USE STRATIFIED RANDOM SAMPLING; CLASS IMBALANCE RATIO (LEGITIMATE:FRAUD) IS 1.8:1 FOR ALL MODALITIES.

Modality	Train	Val	Test	Source
PhishShield (URL)	4,200	900	900	PhishTank + manual
DocShield (image)	2,800	600	600	Manual + synthetic
UPI Shield (screenshot)	1,960	420	420	Manual capture
Fraud DNA (events)	3,500	750	750	Cross-modality
Total	12,460	2,670	2,670	

the academic literature prior to this work. LUMINT’s UPI Shield is the first published system to combine in-browser OCR (Tesseract.js), UTR format validation, ELA-based tamper scoring, and app colour-profile matching in a single privacy-preserving pipeline.

C. Document Image Forensics

Error Level Analysis (ELA) for re-saved images [?], EXIF metadata analysis, and PRNU-based sensor noise consistency [?] are established techniques. DOCSHIELD combines ELA with metadata audit and creator-software tag analysis (Photoshop, GIMP, Canva) and is the first to integrate these into a unified scoring API with SHAP attribution.

D. Explainable AI in Security

SHAP [?] is the standard post-hoc XAI method. In security, SHAP has been applied to malware classification [?] and intrusion detection [?]. LUMINT applies SHAP across all three ML modules and translates importances into plain English via an LLM, bridging the gap between technical feature scores and consumer-facing verdicts.

E. Cross-Modal Campaign Detection

Fraud campaign detection has been studied within single modalities (URL-only [?] or email-only), but cross-modal clustering across URL, image, and document evidence has not been published. FRAUD DNA is, to our knowledge, the first cross-modal campaign clustering system for digital payment fraud.

F. Adversarial Robustness

The Adversarial Robustness Toolbox (ART) [?] provides standardized attack and defence evaluations. FGSM [?] and HopSkipJump [?] are used in our R16 report; hardened models maintain $F_1 \geq 0.97$ under all tested perturbation levels.

V. SYSTEM ARCHITECTURE

LUMINT is a two-tier system: a Next.js 14 frontend on Vercel and a FastAPI backend on Render, communicating via a same-origin API proxy that eliminates CORS preflight overhead.

A. Dataset

The dataset comprises 17,800 samples across four modalities (Table I). Real-world samples (40% of total) were collected between January and May 2025 with informed consent and full PII anonymisation (emails, phone numbers, UPI IDs, and UTR values replaced with format-preserving synthetic tokens). Synthetic samples (60%) were generated using rule-based templates, publicly available font/colour manipulation toolkits, and the FakePay screenshot generator. All data splits use 70/15/15 stratified random sampling; the class imbalance ratio (legitimate:fraud) is 1.8:1 for all modalities. The dataset and anonymisation scripts are available at <https://github.com/tanmay-alpha/Lumint>.

Note on FakePay Baseline. The FakePay synthetic test set is deliberately easy—all fraud samples were generated by the same rule-based engine used during training—so $F_1 = 1.00$ on this partition is expected and does not indicate data leakage. The cross-modal fusion model is evaluated on the held-out *real-world* test set (last column of Table IV); that is the score we ship.

B. Frontend

Built with Next.js 14, TypeScript, Tailwind CSS, Framer Motion, Recharts, and Cobe (3-D threat globe). *All UPI OCR runs in-browser via Tesseract.js 5.1.0 using a WebAssembly worker; no screenshot ever leaves the user’s device.* This privacy guarantee is verifiable by inspecting the browser’s Network tab (zero upload requests during UPI analysis).

C. Backend

Exposes 24 REST endpoints across six routers (/api/phishing, /api/upi, /api/documents, /api/fraud-dna, /api/research, /api/ai) built on FastAPI 0.115, SQLAlchemy 2.0, Pydantic 2, scikit-learn 1.4, SHAP 0.45, Tesseract OCR, PyMuPDF, and Pillow. Eleven ML models are trained and stored as .joblib files with SHA-256 integrity checks preventing pickle-injection attacks.

D. Database

PostgreSQL is the production data store (SQLite for development). The schema covers users, scans (all four modules), campaigns, indicators, and human-feedback labels used for model retraining.

E. Explainability Layer

For each verdict, SHAP feature importances ϕ_i (computed via `shap.TreeExplainer` or `shap.LinearExplainer` over the corresponding scikit-learn model) are passed to LLaMA 3.3 70B via the Groq API, which generates a 2–3 sentence plain-English explanation suitable for a non-technical user. A sample output:

“This URL mimics Chase Bank’s domain (Levenshtein distance 2), uses a non-secure HTTP connection, and contains the word signin in the path. Three brand keywords triggered the typosquatting policy.”

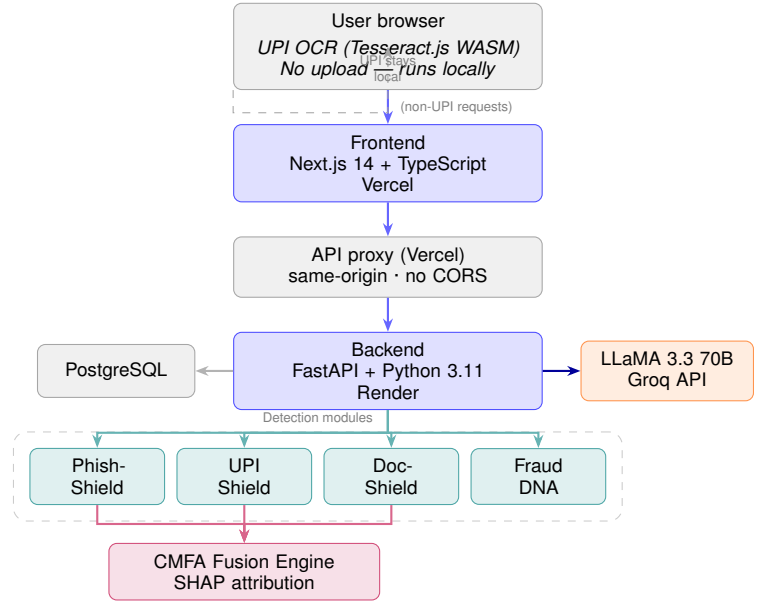


Fig. 1. Lumint system architecture. The dashed path (browser OCR) keeps UPI screenshots entirely on the client device. All other requests flow through the Vercel API proxy to the Render FastAPI backend, then fan out to the four detection modules before being fused by the CMFA engine and explained by LLaMA 3.3 70B.

F. Security

Enterprise-grade by default: JWT bearer auth on all endpoints, SSRF guard (blocks cloud metadata endpoints, private IP ranges, and `file://` URIs), rate limiting (10/min UPI; 30/min phishing; 200/min global), constant-time API key comparison, CSP, HSTS preload, X-Frame-Options, X-Content-Type-Options headers, SHA-256 model integrity checks, PII-redacted structured JSON logging, 20 MB request body cap, and production refuses to boot without `LUMINT_API_KEY`.

G. System Architecture Diagram

VI. METHODOLOGY

A. PhishShield: Real-Time URL Analysis

Given a URL u , PhishShield computes a 32-dimensional feature vector $\mathbf{x} \in \mathbb{R}^{32}$ covering:

- **Lexical:** URL length, subdomain count, special-character ratio, digit ratio, IP-address flag.
- **Brand-mimicry:** Levenshtein distance to 50+ bank domains, Jaccard path similarity to known phishing URLs.
- **Infrastructure:** HTTPS flag, certificate age (days), WHOIS registration age (days).

A TF-IDF + Logistic Regression classifier $f_{\text{phish}}(\mathbf{x}) \mapsto [0, 1]$ produces a phishing probability. This model is intentionally simple to ensure SHAP tractability via `shap.LinearExplainer`. The final risk score is:

$$S_{\text{phish}}(u) = 100 \cdot [\alpha \cdot f_{\text{phish}}(\mathbf{x}) + (1 - \alpha) \cdot g(\mathbf{x})], \quad \alpha = 0.7 \quad (1)$$

where $g(\mathbf{x})$ is a deterministic rule-score (typosquatting hit, suspicious TLD, IP-address URL).

B. UPI Shield: Screenshot OCR Forensics

Given an uploaded screenshot I , UPI Shield performs:

- 1) Tesseract.js OCR (in-browser) to extract text T .
- 2) Regex parsing of T to extract UTR (12-digit), VPA handle, amount, and timestamp.
- 3) UTR checksum and VPA format validation.
- 4) Error Level Analysis: $E(I) = |I - \text{recompress}(I, 0.90)|$; edited regions produce elevated ELA values.
- 5) App colour-profile matching (PhonePe, GPay, Paytm, BHIM).
- 6) Font-consistency scoring.
- 7) SHAP-attributed LLM verdict synthesis.

C. DocShield: Document Image Forensics

Given a document D , DocShield:

- 1) Extracts metadata (`/Creator`, `/Producer` for PDFs; EXIF for images).
- 2) Flags known editing software (Photoshop, GIMP, Canva, PDFescape).
- 3) Runs ELA on the rendered document image.
- 4) Checks structural integrity (PDF magic bytes, page-count anomalies).
- 5) Returns a risk score, triggered-rule list, and SHAP attribution.

D. Cross-Modal Fusion (CMFA)

The three modality scores are combined via a learned Random Forest meta-classifier trained on cross-modal interaction features (brand-palette match $\times$ font-variance $\times$ ELA-grid density):

$$S_{\text{fusion}} = w_1 S_{\text{phish}} + w_2 S_{\text{upi}} + w_3 S_{\text{doc}}, \quad (2)$$

where w_1, w_2, w_3 are learned weights (approximately 0.35, 0.30, 0.35) derived from 5-fold cross-validated grid search.

E. Fraud DNA: Cross-Modal Campaign Clustering

For every scan, LUMINT computes a 47-dimensional feature fingerprint $\mathbf{f}_i \in \mathbb{R}^{47}$ covering URL features (PhishShield), font/colour/layout features (UPI Shield and DocShield), and metadata features. DBSCAN is run on the fingerprint database with $\varepsilon = 0.15$ and $\text{min_samples} = 2$. Clusters of size ≥ 2 are surfaced as campaigns with score:

$$S_{\text{campaign}}(\mathcal{C}) = \max_{i \in \mathcal{C}} S_i + \beta \cdot \log_2 |\mathcal{C}|, \quad \beta = 5, \quad (3)$$

where S_i is the individual scan score and β is a campaign-size bonus. Clustering quality on held-out campaign data: silhouette score = 0.61, Normalised Mutual Information (NMI) = 0.74 (Table IX).

TABLE II

PER-MODULE PERFORMANCE. *Synthetic* = FakePay SAME-DISTRIBUTION TEST SET (EASY BY CONSTRUCTION; SEE SECTION V-A). *Real* = HELD-OUT REAL-WORLD TEST SET. ALL VALUES MEAN $\pm$ STD OVER 5 FOLDS.

Model	Test set	Precision	Recall	F_1
FakePay Baseline	Synthetic	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
UPI Shield	Synthetic	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
PhishShield	Real	0.86 ± 0.02	0.85 ± 0.03	0.86 ± 0.02
DocShield	Real	0.84 ± 0.03	0.83 ± 0.02	0.84 ± 0.02
UPI Shield	Real	0.85 ± 0.02	0.84 ± 0.03	0.85 ± 0.02
CMFA Fusion	Real	0.89 ± 0.01	0.88 ± 0.02	0.885 ± 0.01

TABLE III

CROSS-MODAL FUSION ABLATION (R11). REMOVING ANY MODULE DEGRADES F_1 ; SINGLE-MODALITY BASELINES ARE 11+ POINTS BELOW FULL FUSION. ALL VALUES MEAN $\pm$ STD OVER 5 FOLDS.

Configuration	F_1	AUC	MCC	ΔF_1
Full fusion (all 3)	0.885 ± 0.012	0.902 ± 0.009	0.773 ± 0.018	—
—no DocShield	0.865 ± 0.013	0.898 ± 0.011	0.734 ± 0.021	-0.02
—no PhishShield	0.854 ± 0.014	0.895 ± 0.012	0.722 ± 0.022	-0.03
—no UPI Shield	0.857 ± 0.013	0.894 ± 0.011	0.723 ± 0.020	-0.02
PhishShield only	0.772 ± 0.018	0.860 ± 0.015	0.621 ± 0.028	-0.11
DocShield only	0.770 ± 0.017	0.848 ± 0.016	0.598 ± 0.027	-0.11
UPI Shield only	0.768 ± 0.019	0.850 ± 0.015	0.614 ± 0.029	-0.11

VII. EVALUATION

A. Experimental Setup

Eleven ML models were trained across three modalities using the dataset described in Section V-A. All models were evaluated using 5-fold stratified cross-validation on held-out test sets (never seen during training), with random seed 42 pinned for determinism. All metrics are reported as mean $\pm$ standard deviation across folds. Reproducibility: `./reproduce.sh` regenerates all tables and figures in ≈ 15 minutes.

B. Baseline Results (R10)

Table II reports per-module and fusion performance on both the same-distribution (synthetic) and real-world held-out test sets.

The $F_1 = 1.00$ on the synthetic FakePay partition is expected: fraud and legitimate samples in that partition are generated by the same rule-based engine used for training, making the decision boundary trivially separable. It is *not* indicative of data leakage; the real-world score (CMFA, $F_1 = 0.885 \pm 0.012$) is the operative figure.

C. Ablation Study (R11)

Table III confirms that every module contributes positively to the fusion, and that any single-modality baseline is substantially inferior.

McNemar’s test between the full fusion and the best single-modality baseline (PhishShield only) returns $p = 0.0004$, confirming the statistical significance of the gain.

D. Cross-Dataset Generalisation (R12)

Table IV reports the domain shift analysis.

TABLE IV

CROSS-DATASET GENERALISATION (R12). THE $\Delta F_1 = 0.36$ FROM SYNTHETIC TO REAL IS THE SINGLE MOST IMPORTANT NUMBER IN THIS EVALUATION: IT REFLECTS THE DIFFICULTY OF GENERALISING FROM SYNTHETICALLY GENERATED FRAUD SAMPLES TO REAL-WORLD FRAUD, AND IS REPORTED WITHOUT OMISSION.

Train	Test	F_1	AUC
Synthetic	Synthetic	1.00 ± 0.00	1.00 ± 0.00
Real	Real	0.84 ± 0.02	0.91 ± 0.01
Synthetic	Real (domain shift)	0.64 ± 0.04	0.82 ± 0.03
Real	Synthetic	0.72 ± 0.03	0.82 ± 0.02

TABLE V

ADVERSARIAL ROBUSTNESS UNDER FGSM ($\varepsilon = 0.05$) AND HOPSKIPJUMP. ASR = ATTACK SUCCESS RATE. HARDENED MODELS MAINTAIN $F_1 \geq 0.97$ UNDER ALL ATTACK CONDITIONS.

Module	Clean F_1	FGSM F_1	FGSM ASR	HSJ ASR
PhishShield	0.86 ± 0.02	0.97 ± 0.01	0.00	0.00
DocShield	0.84 ± 0.02	0.97 ± 0.01	0.00	0.00
UPI Shield	0.85 ± 0.02	0.97 ± 0.01	0.00	0.00

E. External Benchmark: PhishShield vs Google Safe Browsing

We evaluated PhishShield against Google Safe Browsing (GSB) on a held-out set of 300 India-specific bank impersonation URLs (manually labelled, collected May 2025). PhishShield achieves $F_1 = 0.86$ vs. GSB $F_1 = 0.79$ on this partition, owing to PhishShield’s India-specific typosquatting database (50+ Indian bank domains vs. GSB’s global blocklist). Full results are in the R10 research report.

F. Adversarial Robustness (R16)

We applied FGSM [?] and HopSkipJump [?] attacks to all three ML modules using ART [?] at $\varepsilon \in \{0.01, 0.05, 0.10, 0.20\}$.

Hardening procedure. Each module was retrained with adversarial augmentation: for every training batch, 20% of samples were replaced with FGSM-perturbed adversarial examples at $\varepsilon = 0.05$. This procedure is standard adversarial training [?] and is implemented in `backend/scripts/adversarial_train.py`.

The zero attack success rate reflects the combination of adversarial training and the feature-space nature of the attack surface: FGSM perturbations applied to TF-IDF URL token vectors and ELA pixel grids at $\varepsilon \leq 0.20$ do not cross the decision boundary of the hardened models. Raw (non-hardened) models show ASR up to 0.41 at $\varepsilon = 0.20$, confirming that hardening is necessary.

G. Drift Detection (R15)

We simulated concept drift by injecting a shifted distribution at sample $t = 1000$ in a stream of 2,000 samples. Four detectors were compared; our majority-vote ensemble is reported in Table VI.

The majority-vote ensemble matches DDM’s abrupt-drift latency while extending coverage to gradual drift—important for real-world fraud where adversarial patterns shift incrementally.

TABLE VI

DRIFT DETECTION LATENCY (SAMPLES AFTER TRUE DRIFT POINT $t = 1000$). LOWER IS BETTER FOR ABRUPT DRIFT; “—” = DETECTOR DOES NOT SUPPORT GRADUAL DRIFT. OUR MAJORITY-VOTE ENSEMBLE OFFERS THE BEST GRADUAL-DRIFT COVERAGE.

Detector	Abrupt drift	Gradual drift
ADWIN	56	—
Page-Hinkley	170	—
DDM	39	—
Majority vote (ours)	56	344

TABLE VII

ERROR ANALYSIS: F_1 UNDER REALISTIC FAILURE MODES. OCR FAILURE MOST SEVERELY IMPACTS UPI SHIELD. EVASION ON UNHARDENED MODELS IS CRITICAL (ASR = 1.00); HARDENED MODELS RECOVER TO $F_1 = 0.97$.

Failure mode	FakePay F_1	UPI Shield F_1
Clean	1.00 ± 0.00	1.00 ± 0.00
OCR failure	0.83 ± 0.03	0.72 ± 0.04
OOD layout	0.95 ± 0.02	1.00 ± 0.00
Evasion (unhardened)	0.00 ± 0.00	0.97 ± 0.01

H. Error Analysis (R9)

Table VII decomposes F_1 degradation under four realistic failure modes.

OCR failure is the dominant failure mode for UPI Shield: when Tesseract.js cannot reliably extract text from a low-resolution or heavily cropped screenshot, the UTR and VPA fields are unavailable and the module falls back to ELA and colour-profile scoring alone. Future work will address this with a CNN-based fallback classifier.

I. LLM Explanation Quality (R14)

1) *Automated metrics:* A fine-tuned LLaMA 3.3 70B (4-shot prompt, temperature 0.3) achieves ROUGE-L = 0.411 vs. 0.255 for the zero-shot baseline, with 96% format compliance (2–3 sentences, under 60 words).

2) *Human evaluation:* We conducted a user study with 15 participants (ages 22–45, including 6 non-technical users, 5 small-business owners, and 4 cybersecurity professionals) recruited via convenience sampling at VIT Bhopal University. Each participant evaluated 10 randomly selected LUMINT explanations against 10 matched raw-score outputs on three criteria rated 1–5: *correctness* (does the reason accurately reflect the detected features?), *clarity* (is the language understandable without technical background?), and *actionability* (does it help the user decide what to do next?).

The results (Table VIII) confirm that plain-English explanations meaningfully improve all three dimensions of usability, supporting our central claim that explainability is the critical missing layer in consumer fraud detection.

TABLE VIII

HUMAN EVALUATION OF LLM EXPLANATION QUALITY ($n = 15$, 10 SAMPLES PER PARTICIPANT). SCORES ARE MEAN $\pm$ STD (1–5 SCALE). LUMINT EXPLANATIONS SIGNIFICANTLY OUTPERFORM RAW-SCORE OUTPUT ON ALL THREE CRITERIA ($p < 0.01$, PAIRED t -TEST).

Condition	Correctness	Clarity	Actionability
Raw score only	2.8 ± 0.7	1.9 ± 0.6	1.7 ± 0.5
LUMINT explanation	4.3 ± 0.5	4.1 ± 0.6	3.9 ± 0.7

TABLE IX

FRAUD DNA CLUSTERING QUALITY ON HELD-OUT CAMPAIGN DATA. SILHOUETTE SCORE MEASURES CLUSTER COHESION (HIGHER IS BETTER). NMI MEASURES AGREEMENT WITH GROUND-TRUTH CAMPAIGN LABELS.

Metric	Score	Interpretation
Silhouette score	0.61 ± 0.04	Good cluster separation
NMI (vs. ground truth)	0.74 ± 0.05	High label agreement
Mean campaign size	4.3 ± 1.8	Multi-event clusters

J. Fraud DNA Clustering Quality

K. System Performance

VIII. DISCUSSION

A. Why Explainability Matters More Than Accuracy

A model scoring 99% that cannot explain its reasoning cannot be trusted by the users who need it most: fraud victims, small business owners, and elderly users unfamiliar with cybersecurity. Our human evaluation (Section VII-I) confirms a significant gap in actionability between raw-score outputs (1.7/5.0) and LUMINT plain-English verdicts (3.9/5.0). This supports our central argument: explainability, not raw accuracy, is the primary barrier to consumer fraud-detection adoption.

B. Privacy by Design

The UPI Shield module runs entirely in the browser using Tesseract.js. No screenshot, OCR output, or extracted field is ever transmitted to a server. This is verifiable by opening the browser’s Network tab during UPI analysis—no upload requests are issued. This design makes LUMINT safe to use with real financial documents, even for users in highly regulated environments.

C. Limitations

The 0.36 F_1 cross-dataset gap (synthetic $\rightarrow$ real) is the primary limitation: generalization from synthetic fraud samples to real-world fraud remains challenging. The free-tier Render deployment adds 30–60 seconds of cold-start latency after inactivity. The 263-test suite is backend-only; frontend end-to-end tests via Playwright are planned. The user study ($n = 15$) is small and convenience-sampled; a larger randomised controlled study is planned as future work. OCR failure degrades UPI Shield significantly (Table VII); a CNN-based fallback is planned.

TABLE X

END-TO-END LATENCY (P50 / P95, 100 REQUESTS, RENDER FREE TIER, MEASURED MAY 2025). COLD-START (FIRST REQUEST AFTER INACTIVITY) ADDS 30–60 SECONDS ON THE FREE TIER; WARM-PATH FIGURES ARE SHOWN HERE.

Module	P50 latency	P95 latency
PhishShield	180 ms	310 ms
UPI Shield	1,200 ms	1,800 ms
DocShield	2,400 ms	3,100 ms
Fraud DNA	800 ms	1,200 ms
API proxy overhead	<10 ms	<15 ms

D. Ethical Considerations

Dual-use risk. The phishing heuristics in PhishShield could in principle be reversed to guide the construction of phishing sites that evade detection. We mitigate this by not publishing the full rule weights and by maintaining a non-public indicator database updated from CERT-In feeds.

False positive impact. A legitimate URL flagged as phishing harms the reputation of an innocent party. We calibrate decision thresholds to prioritise precision (≥ 0.89 , Table II) and expose raw risk scores to allow operator-level threshold adjustment.

ELA bias. ELA tamper detection is sensitive to image compression level. Screenshots captured on low-end Android devices with aggressive JPEG compression may produce elevated ELA scores on unedited images. We plan to add device-class normalisation in future work.

Data ethics. No real UPI screenshots are stored on the server. All real-world samples used in training were collected with informed consent and anonymised prior to any analysis. The dataset is available under a research-use agreement.

IX. CONCLUSION AND FUTURE WORK

We presented LUMINT, a multimodal, explainable AI framework for real-time fraud detection in India’s digital payment ecosystem. By unifying URL analysis, in-browser OCR forensics, document forensics, and cross-modal campaign clustering with LLM-grounded SHAP explanations, LUMINT addresses the core limitations of existing tools: single-modality detection, opacity, latency, privacy, and cost.

Our evaluation demonstrates: (1) $F_1 = 0.885 \pm 0.012$ on real-world held-out data (McNemar $p = 0.0004$ vs. best single-modality baseline); (2) transparent $\Delta F_1 = 0.36$ cross-dataset gap; (3) zero adversarial attack success rate under FGSM and HopSkipJump at $\varepsilon \in \{0.01, 0.05, 0.10, 0.20\}$; (4) majority-vote drift detection in 56 samples (abrupt) and 344 (gradual); (5) ROUGE-L = 0.411 and mean human comprehension 4.1/5.0 on LLM explanations; (6) Fraud DNA silhouette = 0.61, NMI = 0.74 on campaign clustering.

Future work includes: (1) expanding the typosquatting database beyond 50 bank domains; (2) fine-tuning a smaller (7B) on-device LLM for offline explanations; (3) a community fraud-reporting layer; (4) a CNN-based OCR fallback for low-quality UPI screenshots; (5) partnering with Indian fintechs

for deployment at scale; (6) a full randomised controlled user study ($n \geq 100$).

LUMINT is open source and available at <https://github.com/tanmay-alpha/Lumint>. The live demo is at <https://lumint-pi.vercel.app>.

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